

Programming Problems

1. Write a program that has a class `Point` with attributes as the X and Y coordinates. Make two objects of this class and find the midpoint of both the points.
2. Write a program that has a class `Cars`. Create two objects and set `car1` to be a red convertible with price ₦10 thousand and name Pugo, and `car2` to be a blue sedan named Mavo worth ₦76 thousand.
3. Write a program that uses a class attribute to define some default titles for faculty in a college. Display the name along with title and department of the college.
4. Add a method `reflect_x` to class `Point`, which returns a new point which is the reflection of the point about the x-axis. For example, `Point (7, 8)`. `reflect_x` is `Point (7, -8)`.
5. Write a static method that checks whether all words in a list starts with a vowel.
6. Make a class `triangle`. Enter its three sides and calculate its area.
7. Write a menu driven program to read, display, simplify, add, and subtract two fractions.
8. Write a menu driven program to read, display, add and subtract two complex numbers.
9. Write a menu driven program to read, display, add, and subtract two distances.
10. Write a menu driven program to read, display, add, and subtract two time objects.
11. Write a menu driven program to read, display, add, and subtract two height objects.
12. Write a program to read two POINTS and calculate the distance between them.
13. Write a class that has a list of integers as data members and `read()`, `display()`, `find_largest()`, `find_smallest()`, `sum()`, and `find_mean()` as its member functions
14. Make a class `Book` with members, title, author, publisher, and ISBN number. The functions of the class should read and display the data.
15. Write a program that swaps two members of a class.
16. Write a program to find mean of two numbers belonging to two different objects of the same class.
17. Write a program that has a class `student` with data members-roll no and marks in three subjects. Make at least four objects of this class. Use one or more functions that finds total of each student and then sorts the student's records in descending order based on their marks.
18. Write a menu driven program to read, add, and subtract two polynomials.
19. Write a program that uses a time structure within a class. Enter any time and your favourite show's time. The program must display how much time is left for it to start.
20. Write a menu driven program to add or delete items from your inventory of stationary items. You can use a dictionary to store item and the brand.

21. Write a menu driven program to read, add, subtract, multiply, divide, and transpose two matrices.
22. Write a program that displays the details of a cricket player. The details must include his name, matches played, run rate, wickets taken, maiden overs, overs played, number of centuries, and half centuries, etc.